

Solutions about ordinary Points

Suppose $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$

$$\Rightarrow y'' + \frac{a_1(x)}{a_2(x)}y' + \frac{a_0(x)}{a_2(x)}y = 0$$

• divide everything by $a_2(x)$

$$\Rightarrow y'' + p(x)y' + q(x)y = 0 \quad \bullet \text{ standard form}$$

Notes

- 1) if $a_2(x_0) = 0$, then x_0 is singular
- 2) if $a_2(x_0) \neq 0$, then x_0 is ordinary
- 3) if x_0 is ordinary, we can find two linearly individual solutions in the form of a power series, that converges for $|x - x_0| < R$

R = min radius of convergence

= distance between ordinary points and closest singular point

★ $a_2(x)$ is in front of y'' (this is assuming its written in standard form)

ex #1:

Find the min radius of convergence about the ordinary point, $x=7$

$$(x^2-9)y'' + x^2y' + y = 0$$

↑
 $a_2(x)$

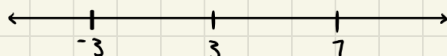
you begin by identifying $a_2(x)$ and setting it to zero and solve for x :

$$a_2(x) = 0$$

$$(x^2-9) = 0$$

$$x = \pm 3$$

← Singular Points



once you have gotten the x -values we will draw a number line and graph the x -values we got with the ordinary point from the two x value we want to identify which of the two is closer to 7 or the ordinary point so:

3 is closet to 7

$$R = |3-7| = \boxed{4}$$

once we identify the number closest to 7 we subtract that number and the ordinary point to find the min radius R

ex #2 $(x^2 - 2x + 26)y'' + xy' - 4y = 0$, $x=0$, $x=1$ (ordinary points)

$$a_2(x) = 0$$

$$x^2 - 2x + 26 = 0$$

$$a = 1$$

$$b = -2$$

$$c = -26$$

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \rightarrow \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(26)}}{2(1)} \rightarrow \frac{2 \pm \sqrt{-100}}{2}$$

$$\frac{2 \pm 10i}{2} \rightarrow 1 \pm 5i$$

• when you are dealing with this all you would have to do is take the $a+bi$ and subtract it from the ordinary point then plug into the square root

★ these are complex conjugates, so they are the same distance from any real point

$$x=0:$$

$$R = |1 + 5i - 0| = \sqrt{(1)^2 + (5)^2} = \boxed{\sqrt{26}}$$

you could have used $1-5i$ and that would be fine

$$x=1:$$

$$|1 + 5i - 1| = |5i| = \boxed{5}$$

Find a Power series Solution, ordinary Point, $x=0$ For:

$$y'' + y = 0, \quad y(0) = 4, \quad y'(0) = 2$$

We will begin by taking the derivative of this formula this formula will be used for every problem:

$$y = \sum_{n=0}^{\infty} C_n x^n$$

You will take the derivative of this equation and plug it into the given problem:

$$y' = \sum_{n=1}^{\infty} C_n n x^{n-1}, \quad y'' = \sum_{n=2}^{\infty} C_n n(n-1) x^{n-2}$$

• For every derivative you take you will increase n by one

now take the derivatives and plug them into the given equation:

$$\sum_{n=2}^{\infty} C_n n(n-1) x^{n-2} + \sum_{n=0}^{\infty} C_n x^n = 0$$

$$k = n-2$$

$$n = k+2$$

$$k = n$$

$$n = k$$

$$\sum_{k=0}^{\infty} C_{k+2} (k+2)(k+1) x^k + \sum_{k=0}^{\infty} C_k x^k = 0$$

$$\sum_{k=0}^{\infty} \left[C_{k+2} (k+2)(k+1) + C_k \right] x^k$$

Now take the highlighted part and isolate for C_{k+2} :

$$C_{k+2} = \frac{-C_k}{(k+2)(k+1)}, \text{ for } k=0, 1, 2, 3, \dots$$

if $k=0 \Rightarrow C_2 = \frac{-C_0}{(2)(1)} = -\frac{C_0}{2!}$ • the 2 in the denominator is the same as 2!

if $k=1 \Rightarrow C_3 = \frac{-C_1}{(3)(2)} = -\frac{C_1}{3!}$ • the 6 in the denominator is the same as 3!

if $k=2 \Rightarrow C_4 = \frac{-C_2}{(4)(3)} = -\left(\frac{-C_0}{2 \cdot 1}\right) \left(\frac{1}{(4)(3)}\right) = \frac{C_0}{4!}$

↑
we know what
 C_2 is equal to
expand 2!

if $k=3 \Rightarrow C_5 = \frac{-C_3}{(5)(4)} = -\left(\frac{-C_1}{3 \cdot 2 \cdot 1}\right) \left(\frac{1}{20}\right) = \frac{C_1}{5!}$

↑
 C_3 is equal to
this we also need
to expand 3!

if $k=4 \Rightarrow C_6 = \frac{-C_4}{6 \cdot 5} = -\left(\frac{C_0}{4 \cdot 3 \cdot 2 \cdot 1}\right) \left(\frac{1}{30}\right) = -\frac{C_0}{6!}$

$$y = C_0 + C_1 x + C_2 x^2 + C_3 x^3 + C_4 x^4 + C_5 x^5 + C_6 x^6$$

$$y = C_0 + C_1 x - \frac{C_0}{2!} x^2 - \frac{C_1}{3!} x^3 + \frac{C_0}{4!} x^4 + \frac{C_1}{5!} x^5$$

• now separate in c_0 and c_1 or get like term together and factor:

$$y = c_0 \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \dots \right) + c_1 \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} + \dots \right)$$

★ as long as you are using power series centered at $x=0$ this statement will be true:

$$c_0 = y(0), \quad c_1 = y'(0)$$

So:

$$c_0 = y(0) = 4 \quad \text{and} \quad c_1 = y'(0) = 2$$

$$y = 4 \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \dots \right) + 2 \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} + \dots \right)$$

ex #2:

Find the Power Series solution, ordinary Point $x=0$:

$$y'' + 2xy' + 2y = 0$$

$$y = \sum_{n=0}^{\infty} C_n x^n$$

$$y' = \sum_{n=1}^{\infty} C_n n x^{n-1}$$

$$y'' = \sum_{n=2}^{\infty} C_n n (n-1) x^{n-2}$$

$$\sum_{n=2}^{\infty} C_n n (n-1) x^{n-2} + 2x \sum_{n=1}^{\infty} C_n n x^{n-1} + 2 \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=2}^{\infty} C_n n (n-1) x^{n-2} + 2 \sum_{n=1}^{\infty} C_n n x^n + 2 \sum_{n=0}^{\infty} C_n x^n = 0$$

$$k = n-2$$

$$n = k+2$$

$$k = n$$

$$n = k$$

$$k = n$$

$$n = k$$

$$\sum_{k=0}^{\infty} C_{k+2} (k+2)(k+1) x^k + 2 \sum_{k=1}^{\infty} C_k k x^k + 2 \sum_{k=0}^{\infty} C_k x^k = 0$$

$$\text{if } k=0$$

$$\Rightarrow C_2 (2)(1) x^0 + 2 [C_0 x^0]$$

$$= 2C_2 + 2C_0$$

$$2C_2 + 2C_0 + \sum_{k=1}^{\infty} C_{k+2} (k+2)(k+1) x^k + 2 \sum_{k=1}^{\infty} C_k k x^k + 2 \sum_{k=1}^{\infty} C_k x^k = 0$$

$$2c_2 + 2c_0 + \sum_{k=1}^{\infty} [c_{k+2} (k+2)(k+1) + 2c_k k + 2c_k] x^k = 0$$

$$\#1) c_{k+2} (k+2)(k+1) + 2c_k k + 2c_k = 0$$

$$\#2) 2c_2 + 2c_0 = 0$$

Solve for c_{k+2} :

$$c_{k+2} = \frac{-2c_k k - 2c_k}{(k+2)(k+1)} = \frac{-2c_k \cancel{(k+1)}}{(k+2) \cancel{(k+1)}} \Rightarrow c_{k+2} = \frac{-2c_k}{(k+2)}$$

$$c_2 = -c_0$$

$$c_{k+2} = \frac{-2c_k}{(k+2)} \quad \text{and} \quad c_2 = -c_0$$

when you have two equations like this you need to make this assumption this will always be the case so this assumption will be valid as long as you have two equations:

$$\text{assume } \left\{ \begin{array}{l} c_0 = 1, \quad c_1 = 0 \\ c_0 = 0, \quad c_1 = 1 \end{array} \right\}$$

So you will begin by with these first:

$$c_0 = 1, \quad c_1 = 0$$

so c_0 and c_1 is already given to us by the assumption we made

$$C_0 = 1, C_1 = 0, C_2 = -1$$

$$\text{if } n=1 \Rightarrow C_3 = \frac{-2C_1}{3} = (0) \left(\frac{-2}{3} \right) = 0$$

• we know what C_1 is which is zero

$$\text{if } n=2 \Rightarrow C_4 = \frac{-2C_2}{4} = (-1) \left(\frac{-2}{4} \right) = 1/2$$

$$\text{if } n=3 \Rightarrow C_5 = \frac{-2C_3}{5} = (0) \left(\frac{-2}{5} \right) = 0$$

$$\text{if } n=4 \Rightarrow C_6 = \frac{-2C_4}{6} = (1/2) \left(\frac{-2}{6} \right) = -1/6$$

equation #1 using $C_0=1, C_1=0$

$$y = C_0 + C_1x + C_2x^2 + C_3x^3 + C_4x^4 + C_5x^5 + C_6x^6$$

$$y = 1 + 0x - x^2 + 0x^3 + 1/2x^4 + 0x^5 - 1/6x^6$$

$$y = 1 - x^2 + 1/2x^4 - 1/6x^6, C_0=1, C_1=0$$

now you have to do the same exact thing but using the other set or, $C_0=0, C_1=1$

$$c_0 = 0, \quad c_1 = 1, \quad c_2 = 0$$

$$\text{if } n=1 \Rightarrow c_3 = \frac{-2c_1}{3} \Rightarrow (1) \left(-\frac{2}{3}\right) = -\frac{2}{3}$$

$$\text{if } n=2 \Rightarrow c_4 = \frac{-2c_2}{4} \Rightarrow (0) \left(-\frac{2}{4}\right) = 0$$

$$\text{if } n=3 \Rightarrow c_5 = \frac{-2c_3}{5} \Rightarrow \frac{-2}{5} \left(-\frac{2}{3}\right) = \frac{4}{15}$$

$$\text{if } n=4 \Rightarrow c_6 = \frac{-2c_4}{6} \Rightarrow \frac{-2}{6} (0) = 0$$

equation #2 using $c_0 = 0, c_1 = 1$

$$y = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + c_5 x^5 + c_6 x^6$$

$$y = 0 + x + 0x^2 - \frac{2}{3}x^3 + 0x^4 + \frac{4}{15}x^5 + 0x^6$$

$$y = x - \frac{2}{3}x^3 + \frac{4}{15}x^5, \quad c_0 = 0, \quad c_1 = 1$$

Find two Power Series Solutions of the given differential equation about the ordinary point $x=0$

$$y'' - 5xy' + y = 0$$

$$y = \sum_{n=0}^{\infty} c_n x^n$$

$$y' = \sum_{n=1}^{\infty} c_n n x^{n-1}, \quad y'' = \sum_{n=2}^{\infty} c_n n(n-1) x^{n-2}$$

$$\sum_{n=2}^{\infty} c_n n(n-1) x^{n-2} - 5x \sum_{n=1}^{\infty} c_n n x^{n-1} + \sum_{n=0}^{\infty} c_n x^n = 0$$

$$\sum_{n=2}^{\infty} c_n n(n-1) x^{n-2} - 5 \sum_{n=1}^{\infty} c_n n x^n + \sum_{n=0}^{\infty} c_n x^n = 0$$

$$k = n-2$$

$$n = k+2$$

$$k = n$$

$$n = k$$

$$k = n$$

$$n = k$$

$$\sum_{k=0}^{\infty} c_{k+2} (k+2)(k+1) x^k - 5 \sum_{k=1}^{\infty} c_k k x^k + \sum_{k=0}^{\infty} c_k x^k = 0$$

$$\text{if } k=0$$

$$\Rightarrow c_2(2)(1)x^0 + c_0 x^0$$

$$\Rightarrow 2c_2 + c_0$$

$$2c_2 + c_0 + \sum_{k=1}^{\infty} c_{k+2} (k+2)(k+1) x^k - 5 \sum_{k=1}^{\infty} c_k k x^k + \sum_{k=1}^{\infty} c_k x^k = 0$$

$$2C_2 + C_0 + \sum_{k=1}^{\infty} [C_{k+2} (k+2)(k+1) - 5C_k k + C_k] x^k$$

$$\#1) C_{k+2} (k+2)(k+1) - 5C_k k + C_k = 0$$

$$\#2) 2C_2 + C_0 = 0$$

$$\#1) C_{k+2} = \frac{5C_k k - C_k}{(k+2)(k+1)} \Rightarrow C_{k+2} = \frac{C_k (5k-1)}{(k+2)(k+1)}$$

$$\#2) = C_2 = -\frac{C_0}{2}$$

$$\text{assume : } \left\{ \begin{array}{l} C_0 = 1, C_1 = 0 \\ C_0 = 0, C_1 = 1 \end{array} \right\}$$

$$C_0 = 1, C_1 = 0, C_2 = -1/2$$

$$\text{if } k=1 \Rightarrow C_3 = \frac{C_1 \cdot 2}{3} = (0) \left(\frac{2}{3}\right) = 0$$

$$\text{if } k=2 \Rightarrow C_4 = \frac{C_2 \cdot 3}{4} = \left(-1/2\right) \left(3/4\right) = -3/8$$

$$\text{if } k=3 \Rightarrow C_5 = \frac{C_3 \cdot 7}{10} = (0) (7/10) = 0$$

$$\text{if } k=4 \Rightarrow C_6 = \frac{C_4 \cdot 19}{30} = \left(-3/8\right) \left(19/30\right) = -19/80$$

equation #1, $C_0=1$, $C_1=0$

$$y = C_0 + C_1 x + C_2 x^2 + C_3 x^3 + C_4 x^4 + C_5 x^5 + C_6 x^6$$

$$y = 1 + 0x - \frac{1}{2}x^2 + 0x^3 - \frac{3}{8}x^4 + 0x^5 - \frac{19}{80}x^6$$

$$y = 1 - \frac{1}{2}x^2 - \frac{3}{8}x^4 - \frac{19}{80}x^6, \quad C_0=1, \quad C_1=0$$

$$C_0=0, \quad C_1=1, \quad C_2=0 \qquad C_{k+2} = \frac{C_k (5k-1)}{(k+2)(k+1)}$$

$$\text{if } k=1 \Rightarrow C_3 = \frac{C_1 \cdot 2}{3} = (1) \left(\frac{2}{3}\right) = \frac{2}{3}$$

$$\text{if } k=2 \Rightarrow C_4 = \frac{C_2 \cdot 3}{4} = (0) \left(\frac{3}{4}\right) = 0$$

$$\text{if } k=3 \Rightarrow C_5 = \frac{C_3 \cdot 7}{10} = \left(\frac{2}{3}\right) \left(\frac{7}{10}\right) = \frac{7}{15}$$

$$\text{if } k=4 \Rightarrow C_6 = \frac{C_4 \cdot 19}{30} = (0) \left(\frac{19}{30}\right) = 0$$

$$\text{if } k=5 \Rightarrow C_7 = \frac{C_5 \cdot 4}{7} = \left(\frac{7}{15}\right) \left(\frac{4}{7}\right) = \frac{4}{15}$$

equation #2, $C_0=0$, $C_1=1$

$$y = C_0 + C_1 x + C_2 x^2 + C_3 x^3 + C_4 x^4 + C_5 x^5 + C_6 x^6 + C_7 x^7$$

$$y = 0 + x + 0x^2 + \frac{2}{3}x^3 + 0x^4 + \frac{7}{15}x^5 + 0x^6 + \frac{4}{15}x^7$$

$$y = x + \frac{2}{3}x^3 + \frac{7}{15}x^5 + \frac{4}{15}x^7, \quad C_0=0, \quad C_1=1$$

Use the Power Series method to solve the given IVP:

$$y'' - 2xy' + 8y = 0, \quad y(0) = 6, \quad y'(0) = 0$$

$$y = \sum_{n=0}^{\infty} c_n x^n$$

$$y' = \sum_{n=1}^{\infty} c_n n x^{n-1}, \quad y'' = \sum_{n=2}^{\infty} c_n n(n-1) x^{n-2}$$

$$\sum_{n=2}^{\infty} c_n n(n-1) x^{n-2} - 2x \sum_{n=1}^{\infty} c_n n x^{n-1} + 8 \sum_{n=0}^{\infty} c_n x^n = 0$$

$$\sum_{n=2}^{\infty} c_n n(n-1) x^{n-2} - 2 \sum_{n=1}^{\infty} c_n n x^n + 8 \sum_{n=0}^{\infty} c_n x^n = 0$$

$$k = n-2$$

$$n = k+2$$

$$k = n$$

$$n = k$$

$$k = n$$

$$n = k$$

$$\sum_{k=0}^{\infty} c_{k+2} (k+2)(k+1) x^k - 2 \sum_{k=1}^{\infty} c_k k x^k + 8 \sum_{k=0}^{\infty} c_k x^k = 0$$

$$\text{if } k=0$$

$$\Rightarrow c_2 (2)(1) x^0 + 8 [c_0 x^0]$$

$$\Rightarrow 2c_2 + 8c_0$$

$$2c_2 + 8c_0 + \sum_{k=1}^{\infty} c_{k+2} (k+2)(k+1) x^k - 2 \sum_{k=1}^{\infty} c_k k x^k + 8 \sum_{k=1}^{\infty} c_k x^k = 0$$

$$2c_2 + 8c_0 + \sum_{k=1}^{\infty} \left[c_{k+2} (k+2)(k+1) - 2c_k k + 8c_k \right] x^k$$

$$\#1) c_{k+2} (k+2)(k+1) - 2c_k k + 8c_k = 0$$

$$\#2) 2c_2 + 8c_0 = 0$$

$$\#1) c_{k+2} = \frac{2c_k k - 8c_k}{(k+2)(k+1)} \Rightarrow c_{k+2} = \frac{c_k (2k - 8)}{(k+2)(k+1)}$$

$$\#2) c_2 = -4c_0$$

$$c_2 = -24$$

$$c_0 = y(0), \quad c_1 = y'(0)$$

$$y(0) = 6, \quad y'(0) = 0$$

$$\text{if } k=1 \Rightarrow c_3 = \frac{-c_1}{2} = (0) \left(-\frac{1}{2} \right) = 0$$

$$\text{if } k=2 \Rightarrow c_4 = \frac{-c_2}{3} = \frac{-(-24)}{3} = 8$$

$$\text{if } k=3 \Rightarrow c_5 = \frac{-c_3}{10} = (0) \left(-\frac{1}{10} \right) = 0$$

$$\text{if } k=4 \Rightarrow c_6 = c_4(0) = 0$$

$$y = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + c_5 x^5 + c_6 x^6$$

$$y = 6 + 0x - 24x^2 + 0x^3 + 8x^4 + 0x^5 + 0x^6$$

$$y = 6 - 24x^2 + 8x^4$$

• as long as you are using a power series centered at $x=0$ this statement is true:

$$c_0 = y(0), \quad c_1 = y'(0)$$